

Paper 4

AI-Assisted Portfolio Triage for Canada's Major Projects: Three-Year Risk Forecasting

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Abstract

Canada's Major Projects Inventory (MPI) provides a national register of large-scale energy, mining, and forestry developments but offers little forward visibility on which proposals will advance. This study applies a dual-model risk engine to the 2024 MPI cohort of 360 active pre-construction projects. A Bayesian likelihood-ratio scorecard, beginning from the historical prior that 28 percent of projects reach construction within five years, updates odds using project attributes such as province, sector, cost, and cleantech status. A Cox proportional hazards model, calibrated on the same cohort, estimates the probability of resolution within three years. Their outputs are blended at a 60–40 weighting to generate two portfolio signals: a three-year build probability and an urgency-adjusted priority index. Results show that 122 projects (34 percent) occupy the high-probability, high-priority quadrant, with Alberta alone contributing 65 of these files. The framework transforms the MPI from a static registry into a dynamic risk dashboard, enabling developers, planners, financiers, and policymakers to triage portfolios, allocate resources, and improve the odds that scarce private and public capital reaches projects most likely to advance.

Introduction

Canada's Major Projects Inventory (MPI) is the most comprehensive public registry of large-scale resource developments, yet it offers little forward guidance on which proposals are likely to break ground. Decision makers face a portfolio of hundreds of projects competing for finite regulatory, financial, and policy attention, without a transparent way to triage which are both feasible and time critical.

Papers 1 and 2 examined the MPI historically through descriptive statistics, cohort tracking, and transition analysis, identifying trends in advancement, attrition, and bottlenecks. Paper 3 introduced a forecasting toolkit that paired two complementary models: a Bayesian likelihood-ratio scorecard, which estimates the probability that a project will advance to construction, and a Cox proportional hazards model, which estimates the expected time to that outcome. Both models were trained and validated on the 2018–2020 MPI cohort, establishing functional accuracy and interpretability.

This paper operationalises those risk engines at the portfolio level. The two models are applied to the 2024 MPI cohort of 360 pre-construction projects, generating three-year build probabilities and an urgency-adjusted priority index for each file. The Bayesian scorecard begins with a historical prior of 28 percent of projects reaching construction within five years and updates those odds with likelihood ratios drawn from project characteristics such as province, sector, cost, and cleantech status. The Cox model contributes a survival-based probability using regression coefficients calibrated on the same cohort. Their outputs are blended at a sixty–forty weight in favour of the Bayesian scorecard, reflecting its stronger near-term discriminatory power, while the Cox model better captures time-to-event dynamics.

By supplying a transparent, algebraic lens on both probability and urgency, this framework transforms the MPI from a static registry into a living dashboard. Developers can benchmark their own projects against sector and regional peers; planners can prioritise reviews within competitive envelopes; financiers can compare risk–reward profiles across subsectors; and policymakers can allocate scarce grant funding toward projects most likely to advance. In short, Paper 4 shifts the focus from model development to portfolio triage, showing how machine-generated insight can augment expert judgment in managing Canada’s pipeline of major projects.

Methods

Dataset and sample construction

The analysis draws on the 2024 edition of Natural Resources Canada’s Major Projects Inventory (MPI), which lists 504 active proposals across Canada’s energy, forestry, and mining sectors. Projects already flagged as Under Construction were deemed successful and removed from the risk pool. This exclusion eliminated 144 records and left a working sample of 360 projects with statuses of Announced & Planning, In Review, or Approved.

Each record included at the 2024 snapshot provides a project identifier, name, sponsor, province or territory, location coordinates, sector and sub-sector grouping, ownership type, estimated capital cost, a cleantech flag, and a count of “reporting years” since the project first appeared in the MPI. These descriptors formed the covariate set.

Data preparation and feature engineering

Categorical fields were cast to strings, and missing values were set to “Unknown” to retain all records. Capital cost was transformed into quintiles (0–4) to prevent extreme values from

dominating the models. Start year was kept as a string. Province, sector, group, start-year, cost quintile, and cleantech were one-hot encoded, along with a province–sector interaction variable to capture location-specific industry effects. The final design matrix contained 86 binary predictors or clues.

Bayesian scorecard

The Bayesian scorecard estimated the probability that a project would advance to construction. It began with the empirical prior that 28 percent of projects in the 2018–2020 MPI cohort had reached *Under Construction* within five years. Each categorical covariate contributed a likelihood ratio (LR). Ratios greater than one increased the odds of success, while ratios less than one reduced them. Sparse categories were stabilised using Laplace smoothing, and unknown values defaulted to a neutral prior (50 percent).

For each project, the active one-hot features were mapped to their likelihood ratios, converted to natural logarithms, and summed. The resulting log-odds were transformed into a posterior probability with the logistic function, yielding p_{bayes} , the project-level probability of advancing to construction.

Cox proportional hazards model

The Cox model estimated the probability of an outcome within three years. Regression coefficients were drawn from the 2018–2020 survival cohort and paired with a baseline survivor function of $S_0(3) = 0.9491$. For each project, a linear predictor was computed as the dot product of the coefficient vector and its one-hot covariate values. Exponentiation produced a hazard

ratio, which was then applied to the baseline survival curve to calculate the three-year probability of exit (either construction or cancellation):

$$p_{cox} = 1 - \text{So}(3)^{\{e^{\{\eta\}}\}}.$$

The same script reported a hazard-based risk score and a years-remaining measure, defined as the historical five-year median time-to-event minus reporting years, floored at 0.25 years to avoid division by zero.

Blending and priority index

To combine the two probability streams, a logit-space weighted average was used with 60 percent weight assigned to the Bayesian scorecard and 40 percent to the Cox model. This yielded the blended three-year build probability, *blended_prob*.

An urgency-adjusted priority index was then defined as the blended probability divided by years remaining. This measure elevated projects nearing their expected decision window, even if their raw probabilities were moderate, and down-weighted long-horizon files.

Quality assurance and assumptions

The scripts included internal checks for missing columns, correct dummy encodings, and probability bounds between zero and one. Coefficient files were hash-logged to ensure reproducibility across runs.

Two modelling assumptions merit emphasis. First, the Bayesian scorecard treated covariates as independent, except for the explicit province–sector interaction. Second, the Cox model inherited the proportional-hazards assumption and a fixed baseline survivor function, calibrated on the

2018–2020 cohort. While recomputing cost quintiles each year maintained relevance, it also introduced potential shifts in comparability across vintages.

Results & Discussion

Portfolio composition

The 2024 MPI dataset contained 360 active pre-construction projects across energy, mining, and forestry. Energy dominated the sample with 231 projects (64 percent), followed by mining with 112 (31 percent) and forestry with 17 (5 percent). Activity was concentrated in British Columbia (99 projects) and Alberta (82), together accounting for more than half of the portfolio. Private sponsors represented 88 percent of proponents, and 64 percent of projects carried a cleantech designation.

Probability distributions

The two risk engines displayed distinct output patterns. Bayesian probabilities (p_{bayes}) spanned the full range, from 0.1 to 99.8 percent, with a mean of 51.9 percent. The Cox model was more conservative, clustering between 0.5 and 8.4 percent with a mean of 3.1 percent. Figure 1 compares these distributions by province. The blended probability ($blended_prob$), weighted 60 percent Bayesian and 40 percent Cox, produced mid-range values from 0.3 to 62.2 percent, with a mean of 32.3 percent.

The urgency-adjusted priority index, defined as blended probability divided by years remaining, was more dispersed. It ranged from 0.001 to 2.43 with a standard deviation of 0.682, highlighting the heterogeneity of timing pressures across the portfolio.

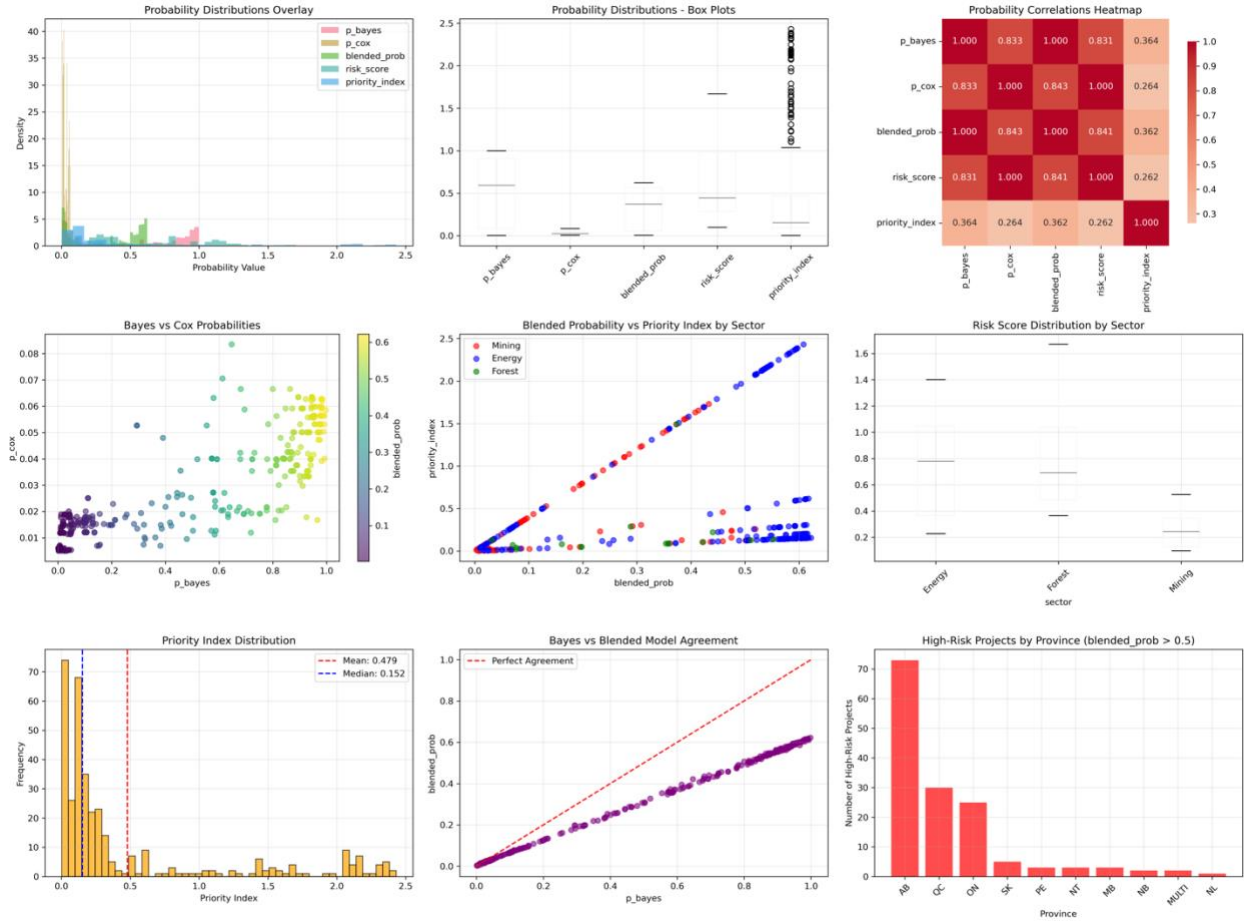


Figure 1: Probability and Priority of MPI 2024 Pre-Construction Projects by Province.

Sectoral and regional patterns

Figure 2 disaggregates results by sector. Energy projects displayed the broadest distribution of Bayesian probabilities, with a distinctive bimodal shape near 0.1 and 0.9, while mining projects clustered close to zero and forestry projects occupied an intermediate range. Sector averages reinforced this contrast. Energy recorded mean values of 0.658 for p_{bayes} , 0.410 for $blended_prob$, and 0.557 for the priority index, all well above mining, which averaged 0.243, 0.151, and 0.362, respectively.

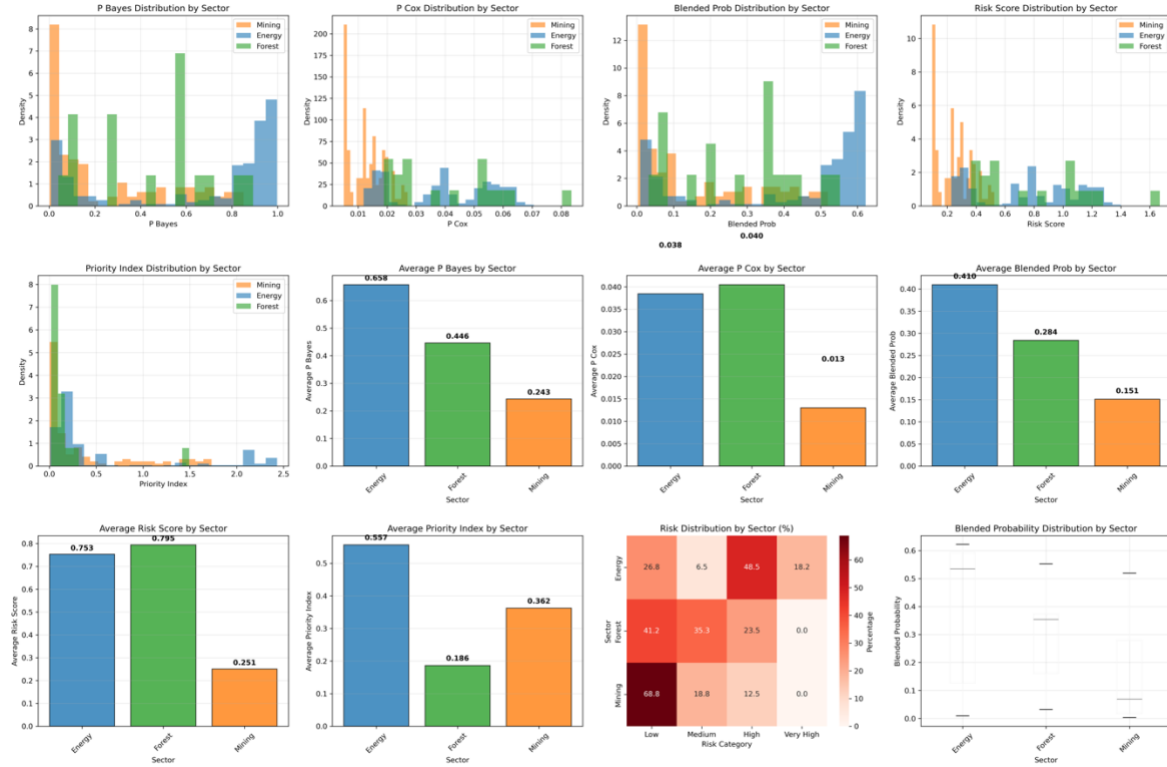


Figure 2: Probability and Priority of MPI 2024 Pre-Construction Projects by Sector.

Hazard-based risk scores showed similar separation. Energy and forestry projects posted average values near 0.75–0.80, three times higher than mining’s 0.25. Figure 3 presents a correlation heat-map of the five metrics, with the strongest links between p_cox and risk score ($\rho = 1.000$) and between p_bayes and $blended_prob$ ($\rho = 0.9998$). Priority index correlations with probability measures were much weaker, in the 0.26–0.36 range, underscoring that urgency captures a distinct dimension.

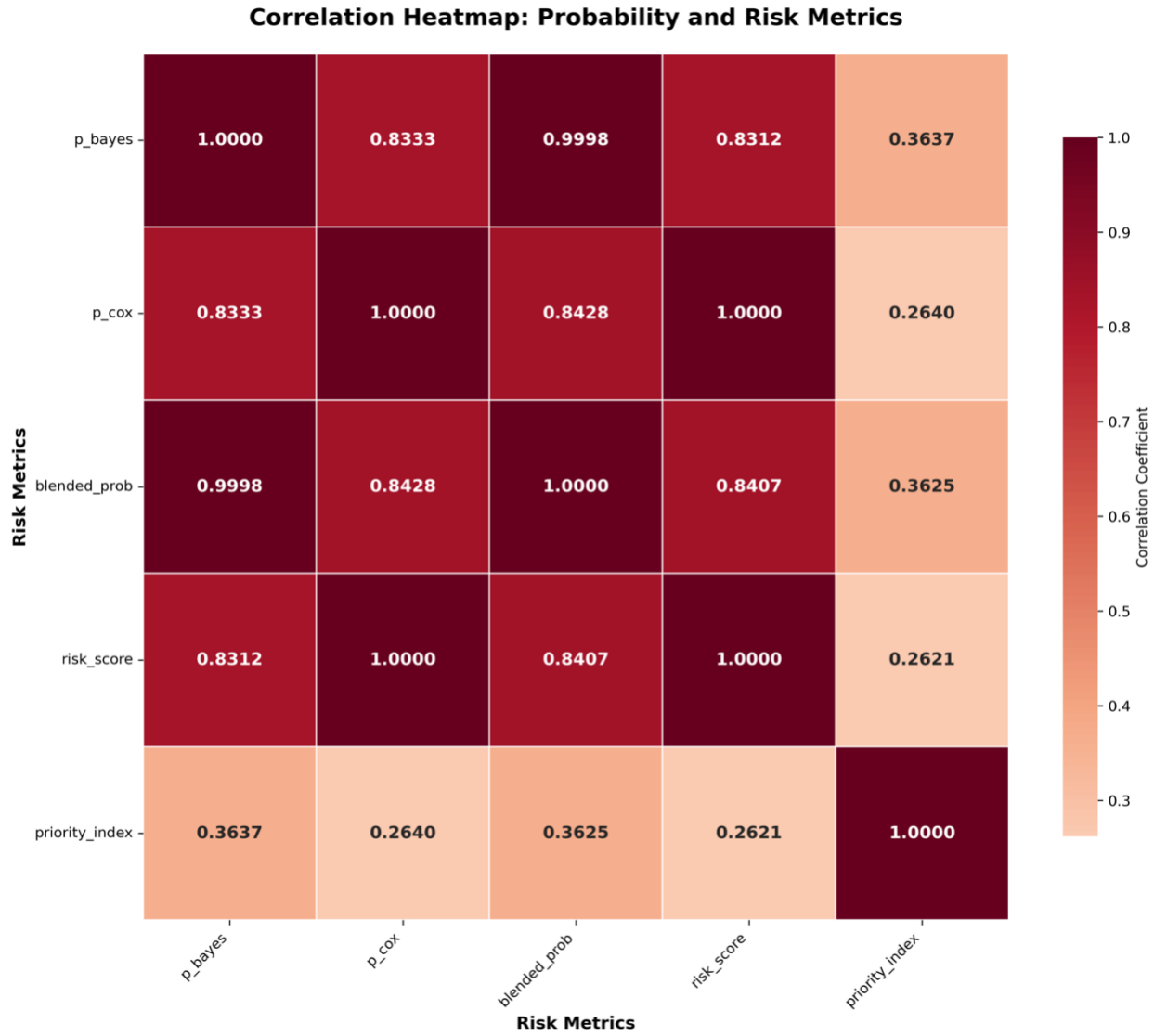


Figure 3: Heat Map of Interrelation of Probability and Risk Metrics.

Triage matrix

The triage framework cross-tabulates blended probability with the priority index, producing four quadrants (Figure 4). The high-probability, high-priority quadrant contained 122 projects, or 34 percent of the portfolio. Energy dominated this group with 107 projects, while mining contributed 13 and forestry two. These files combined strong odds of success with near-term decision pressure.

The low-priority, high-probability quadrant contained 58 projects (16 percent). Their blended probabilities averaged 0.546, but low urgency (priority index 0.137) allowed for longer development windows.

The high-priority, low-probability quadrant contained 59 projects (16 percent), with a mean blended probability of 0.178 and a mean priority index of 0.645. Mining dominated this group, suggesting proposals facing acute timing pressure but low odds of near-term success.

The remaining 121 projects (34 percent) fell into the low-probability, low-priority quadrant. These files had an average blended probability of 0.064 and a mean priority index of 0.053, signalling weak fundamentals and limited urgency.



Figure 4: Project Triage Matrix by Sector.

Regional patterns reinforce these distinctions (Figure 5). Alberta hosted 82 active projects, of which 65—or 79 percent—fell into the high-probability, high-priority quadrant, reflecting a concentrated pipeline of shovel-ready projects. Ontario and Quebec each recorded more than 40 projects but distributed more evenly across quadrants. British Columbia showed a bimodal profile, with one cluster of urgent, higher-risk files and a larger set of speculative, long-horizon proposals.

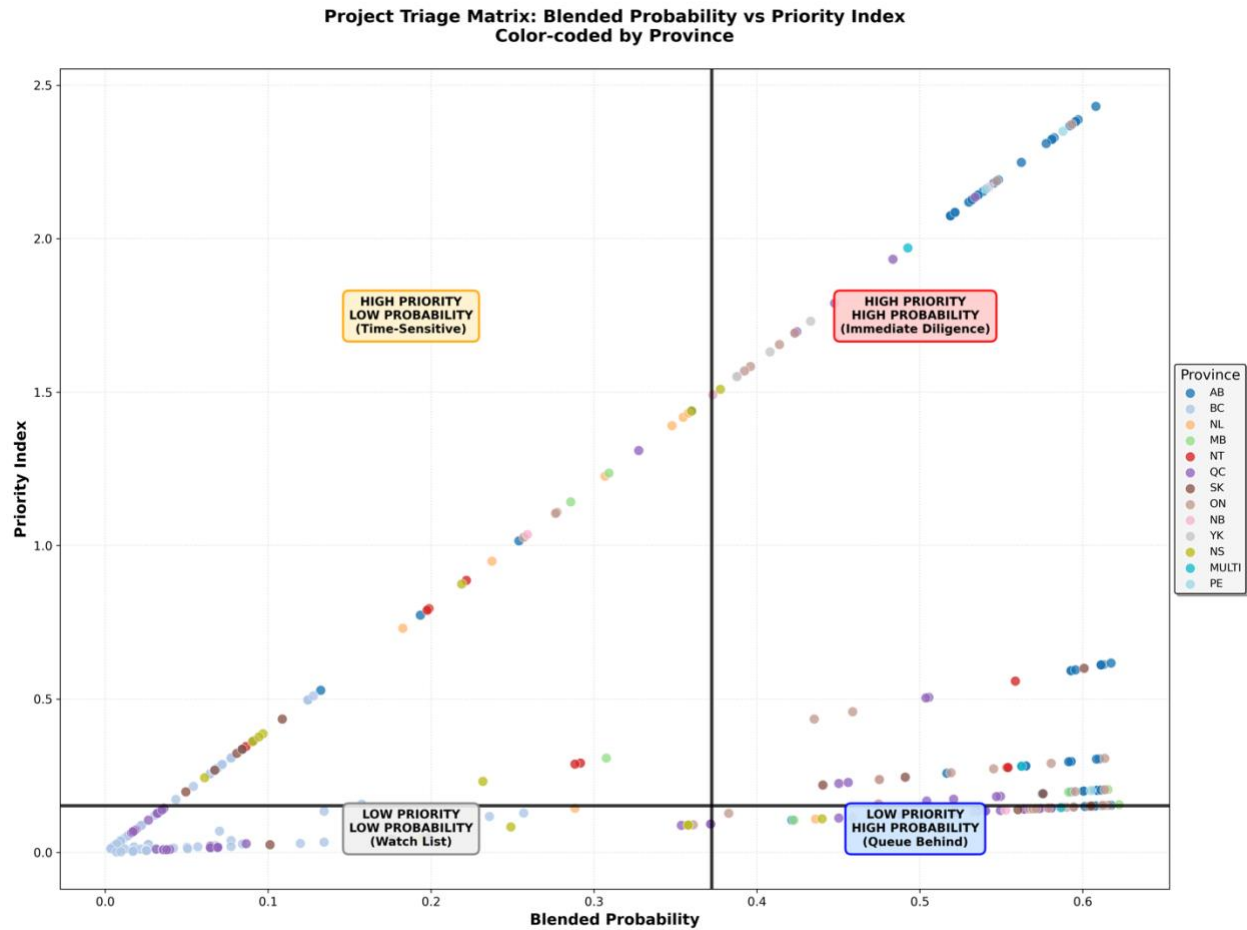


Figure 5: Project Triage Matrix by Province.

Key findings

The blended risk engine produced three consistent themes. First, geography and sector remained the dominant drivers of probability and timing, with Alberta and the energy sector consistently producing higher-probability, higher-urgency clusters. Second, urgency reshaped the portfolio map, lifting a small subset of projects into priority despite moderate raw probabilities. Third, the triage framework identified stark trade-offs: some projects were likely but not urgent, others urgent but unlikely, and many both unlikely and low priority. These distinctions illustrate how probability and urgency must be considered jointly in portfolio management.

Implications for developers

For developers, the blended probability and priority index provide a tactical playbook for sequencing work. Projects in the high-probability, high-priority quadrant signal that every month counts. Accelerating front-end engineering, securing long-lead equipment, and finalizing stakeholder consultations become value-accretive moves. Files in the low-priority, high-probability quadrant, by contrast, have solid fundamentals but more forgiving timelines. Sponsors can use that breathing room to negotiate contracts, secure offtake agreements, or refine financing terms before moving to a final investment decision.

Projects in the high-priority, low-probability quadrant present the most difficult choices. These proposals face acute timing pressure but weak odds of success without intervention. Developers must decide whether to intensify efforts by de-risking, attracting partners, or lobbying for incentives, or to scale back and preserve option value. Proposals in the low-probability, low-priority quadrant often warrant “slow-burn” treatment: minimal spend to preserve rights or stakeholder engagement until a catalyst, such as commodity price changes or new policy, shifts fundamentals.

Implications for planners

For planners, the framework offers a transparent queue for allocating regulatory and analytical bandwidth. High-probability, high-priority projects can be placed in a fast lane, as early approvals are more likely to yield completed builds. Conversely, low-priority, high-probability projects can be advanced at a slower pace without jeopardizing outcomes. The quadrant view therefore allows planners to match review intensity to expected payoff and urgency, improving throughput across the MPI pipeline.

Implications for financiers

For financiers, the triage framework sharpens risk–reward assessment. A 26-point blended-probability spread separates energy projects from mining, implying different hurdle rates and structuring terms. Urgency amplifies these distinctions. Top-decile projects may require higher pricing or contingent capital because delays threaten offtake agreements or regulatory deadlines. Conversely, lower-priority projects can absorb slower financing cycles without compromising value. By anchoring decisions to observable probabilities and urgency, lenders and investors can calibrate terms more directly to project risk.

Implications for policymakers and public funders

For policymakers and public funders, the framework highlights where limited budgets can achieve the greatest impact. High-probability, high-priority clusters, particularly in Alberta and the energy sector, signal where funding support may translate most reliably into construction. The same grid also anticipates workforce needs: multiple projects approaching the critical path in the same jurisdiction point to emerging skilled-labour shortages. Grant programs can incorporate the blended probability and priority index into scoring formulas, generating transparent, data-driven queues for competitive envelopes.

Conclusion

This study applied a dual-model risk framework to Canada’s 2024 Major Projects Inventory, evaluating 360 active pre-construction projects across energy, mining, and forestry. A Bayesian likelihood-ratio scorecard and a Cox proportional hazards model, blended at a 60–40 weighting, generated two complementary signals: (i) a three-year build probability and (ii) an urgency-

adjusted priority index. Together these measures produced a portfolio map that distinguishes projects by both likelihood of construction and time pressure.

The results revealed a skewed landscape. High-probability, high-priority projects accounted for one third of the portfolio, dominated by energy developments in Alberta. At the opposite end, one third of projects fell into the low-probability, low-priority quadrant, suggesting limited prospects for near-term advancement. The remaining quadrants highlighted important trade-offs: projects with strong fundamentals but modest urgency, and projects under acute time pressure but with low odds of success.

For developers, the framework provides a benchmark to diagnose whether weak probability or tight timing is the main drag on a file, guiding investment and sequencing decisions. For planners, it offers a transparent queue to allocate regulatory bandwidth. For financiers, it clarifies risk–reward spreads across sectors and timelines. And for policymakers, it provides a defensible basis for prioritising scarce funding toward clusters of high-probability, high-urgency projects.

Two caveats remain. Static covariates fix conditions at the 2024 snapshot, and the MPI may omit late-breaking proposals. Rolling annual updates, together with qualitative judgment, would mitigate these gaps. Even so, the blended probability and priority index provide an objective first filter that sharpens portfolio triage without replacing detailed project-level diligence.

By translating Canada’s static project registry into a dynamic risk dashboard, this study demonstrates how transparent, auditable models can improve the allocation of attention, capital, and policy support. The framework enables developers, planners, financiers, and funders to focus on projects most likely to advance, thereby increasing the efficiency of both private investment and public programs.